

Item Metadata
(Product Descriptions)

Encode

Sentence Embeddings
(768-dim vectors)

Train FAISS OPQ

Learn: Rotation Matrix R
+ Codebook Centroids C

Static Quantization

Discrete Semantic IDs
(Fixed, Frozen)

Convert to Tokens

Token Sequences
(input_ids)

Forward Pass

GPT2 Backbone
(Transformer)

Prediction Heads

Softmax Over
Token Embeddings

Compute Loss

CrossEntropy Loss

Backprop

Update Model Weights